Exact values

1

a $\frac{\sqrt{3}}{2}$ **b** $\frac{1}{2}$ **c** $\frac{1}{\sqrt{3}}$ **d** $\frac{1}{\sqrt{2}}$

2

a $\theta = 30^\circ$ **b** $\cos\theta = \frac{\sqrt{3}}{2}$

3

a $x = \frac{\sqrt{30}}{2}$ **b** $h = 1$

c $x = 2$ **d** $P = 17$

e $w = 22$ **f** $x = 4\sqrt{6},\ y = 4\sqrt{3}$

g $u = 8,\ v = 4\sqrt{3}$ **h** $a = \frac{9\sqrt{3}}{2},\ b = \frac{9}{2}$

i $x = 7\sqrt{2},\ y = 14\sqrt{2}$ **j** $a = 9\sqrt{2},\ b = 6\sqrt{3},\ c = 12\sqrt{3}$

4

a 45° **b** 60°

5

a $\angle DCB, \angle BAC$ **b** $m = \frac{20\sqrt{21}}{3},\ n = 5\sqrt{21}$

6

a

i $p = \frac{5\sqrt{3}}{3}$ **ii** $q = \frac{10\sqrt{3}}{3}$ **iii** $r = \frac{5}{2}$ **iv** $s = \frac{5\sqrt{3}}{2}$

b $\frac{15\sqrt{3}}{2} + \frac{5}{2}$

7

a $x = 5$ **b** $y = 5\sqrt{3}$ **c** $5\sqrt{3} + 35$

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